



UNIT MATEMATIK
KOLEJ MATRIKULASI PAHANG

MATHEMATICS UNIT
PAHANG MATRICULATION COLLEGE

MODEL PSPM SM015

MATHEMATICS 1

Please do it without being asked !!

ARAHAN KEPADA CALON:

Kertas soalan ini mengandungi 10 soalan.

Jawab *semua* soalan.

Markah penuh yang diperuntukkan bagi tiap-tiap soalan atau bahagian soalan ditunjukkan dalam kurungan pada penghujung soalan atau bahagian soalan.

Semua langkah kerja hendaklah ditunjukkan dengan jelas.

Kalkulator saintifik yang tidak boleh diprogramkan sahaja boleh digunakan.

Jawapan berangka boleh diberi dalam bentuk π , e , surd, pecahan atau sehingga tiga (3) angka bererti, di mana-mana yang sesuai, kecuali jika dinyatakan dalam soalan.

INSTRUCTION TO CANDIDATE:

This question paper consists of **10** questions.

Answer *all* questions.

The full marks for each question or section are shown in the bracket at the end of the question or section.

All steps must be shown clearly.

Only non-programmable scientific calculators can be used.

Numerical answers can be given in the form of π , e , surd, fractions or up to three (3) significant figures, where appropriate, unless stated otherwise in the question.

LIST OF MATHEMATICAL FORMULAE**Trigonometry**

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

Differentiation

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

The Sum Rule

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

Product Rule

$$\frac{d}{dx}(u(x)v(x)) = v \frac{du}{dx} + u \frac{dv}{dx}$$

Quotient Rule

$$\frac{d}{dx} \left(\frac{u(x)}{v(x)} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Chain Rule

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx}$$

Shape	Volume	Surface Area
Sphere	$V = \frac{4}{3} \pi r^3$	$V = 4\pi r^2$
Right circular cone	$V = \frac{1}{3} \pi r^2 h$	$S = \pi r^2 + \pi r h$
Right circular cylinder	$V = \pi r^2 h$	$S = 2\pi r^2 + 2\pi r h$

SECTION A [25 MARKS]

This section consists of **3** questions. Answer **all** questions.

1. Evaluate the following limits:

(a) $\lim_{x \rightarrow -3} \frac{x+3}{x^3+27}$ [4 marks]

(b) $\lim_{x \rightarrow \infty} \sqrt{\frac{5x+7}{6x-5}}$ [3 marks]

(c) $\lim_{x \rightarrow 2^-} \frac{x+1}{x^2-4}$ [2 marks]

2. Find the first derivatives of

(a) (i) $y = 8^{7x+6}$ [2 marks]

(ii) $f(x) = \ln\left(\frac{x^2-1}{\sqrt{x-1}}\right)$ [3 marks]

(b) Find the value of $\frac{dy}{dx}$ for $y = \sin 2x - x \cos x$, when $x = \frac{\pi}{3}$. [4 marks]

3. Given a curve $f(x) = 27 + 9x - 3x^2 - x^3$.

(a) Find the coordinates of the stationary points of the curve. [5 marks]

(b) Determine whether the stationary points are maximum or minimum. [2 marks]

SECTION B [75 MARKS]

This section consists of **7** questions. Answer **all** questions.

1. Given that $z_1 = 3 - 3i$ and $z_2 = 3 + 2i$, where $i = \sqrt{-1}$. Express $\frac{(\overline{z_1}z_2)}{13} + \left(\frac{i^3}{-z_2}\right)$ in the form of $a + bi$, where $a, b \in \mathbb{R}$. [6 marks]
2. (a) Solve the equation $x^{\ln x - 2} = e^3$. [6 marks]
(b) Obtain the solution set for $x - 1 \leq x^2 + 3x \leq x + 3$. [6 marks]
3. Given $f(x) = \frac{x+a}{x-a}$ and $g(x) = bx + 3$, where a and b are constants.
Find the possible values of a and b if $fg(1) = 2$ and $gf(1) = -1$. [6 marks]
4. Given the function $g(x) = \frac{1}{2x-5}$.
(a) Find the domain and range for $g(x)$. [2 marks]
(b) Show that $g(x)$ is a one-to-one function. Hence find $g^{-1}(x)$. [5 marks]
(c) On the same axis, sketch the graph of $g(x)$ and $g^{-1}(x)$. [3 marks]
(d) Show that $g \circ g^{-1}(x) = x$. [2 marks]
5. Functions $f(x) = \begin{cases} x+2, & x \geq 2, \\ x^2 - 4x + a, & x < 2, \end{cases}$ and $g(x) = \begin{cases} \frac{b}{x}, & x \geq 2, \\ 4-x, & x < 2, \end{cases}$ are continuous at $x = 2$.
(a) (i) Find the values of a and b . [4 marks]
(ii) Show that the function $f + g$ is continuous at $x = 2$. [4 marks]
(b) Find the vertical and horizontal asymptote(s) for the function $\frac{\sqrt{x^2+9}}{x}$, if they exist. [6 marks]

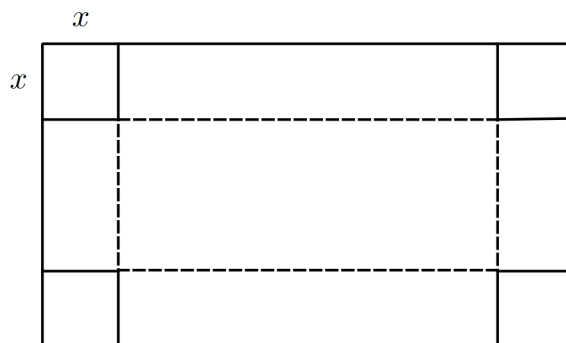
6. (a) A curve is defined by the parametric equations $x = 3t - \frac{1}{t}$ and $y = t + \frac{3}{t}$, where $t \neq 0$.

(i) Show that $\frac{dy}{dx} = \frac{t^2 - 3}{3t^2 + 1}$. [4 marks]

(ii) Find $\frac{d^2y}{dx^2}$. If $t \in \mathbb{Z}$, evaluate $\frac{d^2y}{dx^2}$ when $y = \frac{7}{2}$. [9 marks]

(b) Show that $\frac{dy}{dx}$ can be expressed as $\frac{dy}{dx} = \frac{1}{3} - \frac{10}{3(3t^2 + 1)}$. [2 marks]

7. An open rectangular box is to be made from a piece of paper of width 8 cm and length 15 cm. A square of width x cm is cut from each corner of the paper and the sides are folded up as shown in the figure below. Find the width, length and height of the box that gives the maximum volume.



[10 marks]

END OF QUESTION PAPER

Answers

SECTION A

1. (a) $\frac{1}{27}$ (b) $\frac{\sqrt{30}}{6}$ (c) $-\infty$

2. (a) (i) $\frac{dy}{dx} = 7(\ln 8)8^{7x+6}$ (ii) $f'(x) = \frac{2x}{x^2-1} - \frac{1}{2(x-1)}$ (b) -0.5931

3. (a) $(-3, 0), (1, 32)$ (b) Min: $(-3, 0)$, Max: $(1, 32)$

SECTION B

1. $\frac{5}{13} + \frac{12}{13}i$

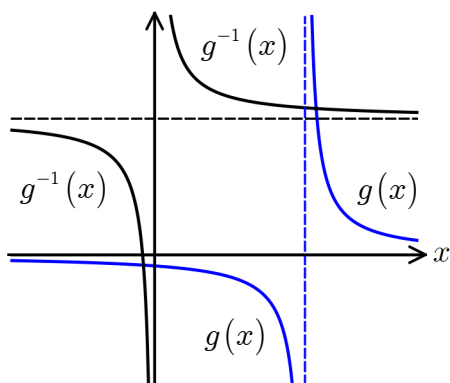
2. (a) $x = e^3, x = e^{-1}$ (b) $\{x : -3 \leq x \leq 1\}$

3. $a = \frac{1}{3}, a = 1, b = -2, b = 0$

4. (a) $D_g = \left(-\infty, \frac{5}{2}\right) \cup \left(\frac{5}{2}, \infty\right), R_g = (-\infty, 0) \cup (0, \infty)$ (b) $DIY, g^{-1}(x) = \frac{1}{2x} + \frac{5}{2}$

(c)

(d) DIY



5. (a) (i) $a = 8, b = 4$ (ii) DIY (b) HA: $y = -1, y = 1$, VA: $x = 0$

6. (i) DIY (ii) $\frac{20t^3}{(3t^2+1)^3}, \frac{160}{2197}$ (b) DIY

7. Width = $\frac{14}{3}$ cm, Length = $\frac{35}{3}$ cm, Height = $\frac{5}{3}$ cm

SECTION A [25 MARKS]

This section consists of **3** questions. Answer **all** questions.

1. Evaluate.

(a) $\lim_{x \rightarrow -\infty} \frac{\sqrt{3x^4 + x}}{x^2 - 6}$ [4 marks]

(b) $\lim_{x \rightarrow -3} \frac{2 - \sqrt{x^2 - 5}}{x + 3}$ [5 marks]

2. (a) If $y = \sin(x^2 + 1)$, show that $x \frac{d^2y}{dx^2} - \frac{dy}{dx} + 4x^3y = 0$. [5 marks]

(b) Find the first derivative of $e^{3xy} = 11x - 5y$. [4 marks]

3. By considering the first and second derivatives of $f(x)$, show that there is only one maximum point on the graph $f(x) = \frac{\ln(x-1)}{x-1}$. [7 marks]

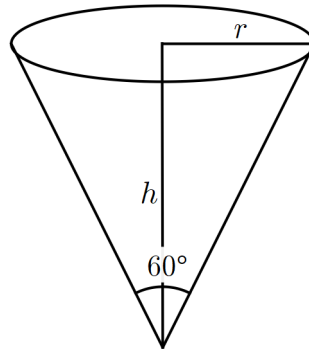
SECTION B [75 MARKS]

This section consists of **7** questions. Answer **all** questions.

1. Solve for p and q , where $p \neq q$, such that $\frac{p+qi}{3i} = 3 + \sqrt{-16} - i^3$. [6 marks]
2. (a) If $\sqrt{7-3\sqrt{5}} = \sqrt{x} - \sqrt{y}$, determine the values of x and y . [6 marks]
(b) Find the solution set for the inequality $\frac{-4-x}{x-3} \geq x+4$. [6 marks]
3. Given $g(x) = \sqrt[3]{x}$ and $h(x) = \frac{1}{x^3}$. Find $f(x)$ such that $(f \circ g \circ h)(x) = \frac{x}{x+1}$. [4 marks]
4. Given that $f(x) = e^{-2x} + 3$ and $(g \circ f)(x) = \frac{1}{2} \ln(e^{-2x} + 2)$.
(a) Find domain and range of $f(x)$. [3 marks]
(b) Find $g(x)$. [5 marks]
(c) Hence, find $g^{-1}(x)$. Sketch the graph of $g(x)$ and $g^{-1}(x)$ on the same axes. [6 marks]
5. Given that $f(x) = \frac{1-x^2}{2x+1}$.
(a) Show that the $f(x)$ has one vertical asymptote and state the equation. [3 marks]
(b) Find the horizontal and vertical asymptotes of the function $g(x) = \frac{1}{f(x)}$. [8 marks]
(c) Sketch the graph of $g(x)$. [3 marks]
6. (a) If $y = e^x \ln(1+x)$, show that $(x+1)^2 \left(\frac{d^2y}{dx^2} - \frac{dy}{dx} \right) = xe^x$. [6 marks]
(b) Determine whether the function $f(x) = \begin{cases} x^2, & x \leq 1, \\ x+1, & x > 1, \end{cases}$ is differentiable at $x = 1$. [5 marks]
(c) Find $\frac{dy}{dx}$ for $y = x^{\sin x}$. [4 marks]

7. (a) Use the derivative to find the maximum area of a rectangle that can be inscribed in a semicircle of radius 10 cm. [5 marks]

(b)



Water drains from the conical tank shown at a rate of 6 cm^3 per second. Find the rate of change in height of the water when the volume of water in the cone is $24\pi \text{ cm}^3$.

[5 marks]

END OF QUESTION PAPER

Answers

SECTION A

1. (a) $\sqrt{3}$ (b) $\frac{3}{2}$

2. (a) *DIY* (b) $\frac{dy}{dx} = \frac{11 - 3ye^{3xy}}{3xe^{3xy} + 5}$

3. *DIY*

SECTION B

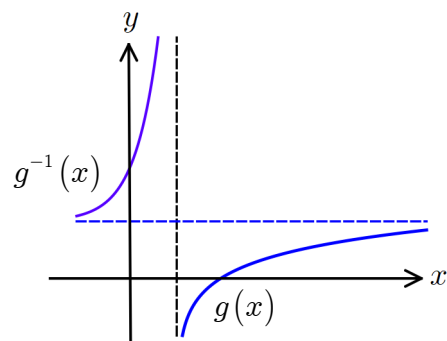
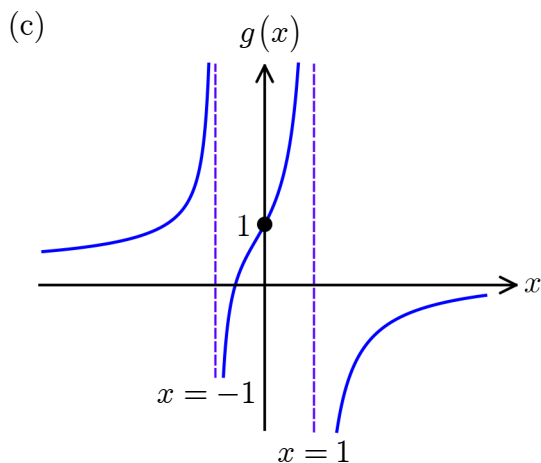
1. $p = -15, q = 9$

2. (a) $x = \frac{9}{2}, y = \frac{5}{2}$ (b) $\{x : x \leq -4 \text{ or } 2 \leq x < 3\}$

3. $f(x) = \frac{1}{1+x}$

4. (a) $D_f = \mathbb{R}, R_f = (3, \infty)$ (b) $g(x) = \frac{1}{2}\ln(x-1)$ (c) $g^{-1}(x) = e^{2x} + 1$

5. (a) VA: $x = -\frac{1}{2}$ (b) HA: $y = 0$, VA: $x = -1, x = 1$



6. (a) *DIY* (b) f is not differentiable at $x = 1$ (c) $\frac{dy}{dx} = x^{\sin x} \left[\sin x \left(\frac{1}{x} \right) + \ln x (\cos x) \right]$

7. (a) 100 cm^2 (b) $\frac{dh}{dt} = -\frac{1}{2\pi} \text{ cms}^{-1}$

SECTION A [25 MARKS]

This section consists of **3** questions. Answer **all** questions.

1. Evaluate the following (if exist):

(a) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{|x - 2|}$ [4 marks]

(b) $\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2 + 3x}}{5x + 1}$ [5 marks]

2. Find $\frac{dy}{dx}$ of the following:

(a) $y = x^{\sec x}$ [5 marks]

(b) $y = \cot \sqrt{4x^2 + 1}$ [4 marks]

3. Given the curve $f(x) = x^3 - \frac{3}{2}x^2 - 18x + 1$. Use the first derivative test to determine the maximum and minimum points of $f(x)$. [7 marks]

SECTION B [75 MARKS]

This section consists of **7** questions. Answer **all** questions.

1. Given $\bar{z} - \frac{1}{\bar{z}} = \frac{8}{5} - \frac{6}{5}i$. Obtain $\left| \bar{z} - \frac{1}{\bar{z}} \right|$ and hence, determine the values of real numbers α and β if $\alpha + \beta i = \left| \bar{z} - \frac{1}{\bar{z}} \right| \left(\bar{z} - \frac{1}{\bar{z}} \right)^2$. [6 marks]

2. (a) Solve the equation $2 + \log_2 x = 15 \log_x 2$. [5 marks]

- (b) Find the interval for x so that $\left| x - \frac{2}{x} \right| \leq 1$. [7 marks]

3. Given that $f(x) = 3x + 8$ and $h(x) = 3x^2 - 12x + 2$ such that $(f \circ g)(x) = h(x)$.

- (a) Determine $g(x)$ in the form $a(x + b)^2 + c$. [5 marks]

- (b) State the domain and range of $g(x)$. [2 marks]

- (c) Sketch $g(x)$. [3 marks]

4. Given $f(x) = \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}$.

- (a) Simplify $f(x)$ and evaluate $f\left(\frac{1}{2}\right)$. [5 marks]

- (b) The domain of $f(x)$ is a set of real numbers except for three specific numbers. Determine what those numbers are. [3 marks]

5. (a) A function $f(x)$ is defined by $f(x) = \begin{cases} a + \frac{1}{x}, & -5 \leq x < -2, \\ x^2 + b, & -2 \leq x < 3, \\ \sqrt{x+6}, & 3 \leq x < 5. \end{cases}$

Determine the values of a and b so that $f(x)$ is continuous in the interval $(-5, 5)$.

[5 marks]

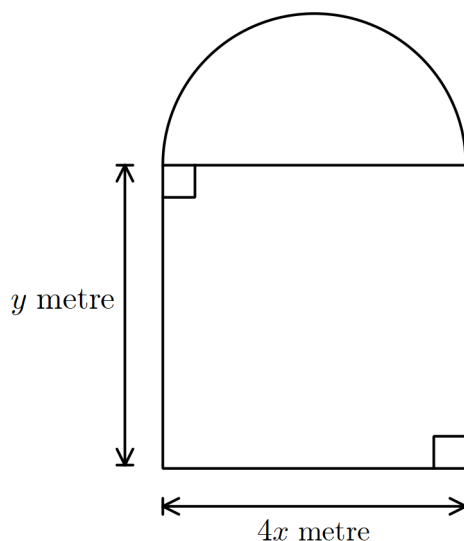
- (b) Given $g(x) = \frac{x}{|x| - 1}$. Determine its vertical and horizontal asymptote. [9 marks]

6. A curve with equation $x^2 - 3y^2 = ae^{y-2x} + by - 6$, where a and b are constants, passes through the point $(1, 2)$.

(a) Given $\frac{dy}{dx} = 1$ at point $(1, 2)$, determine the values of a and b . [9 marks]

(b) Evaluate $\frac{d^2y}{dx^2}$ at $(1, 2)$. [6 marks]

7. Diagram below shows a front view of a window.



The arc of the window shown in the diagram is a semicircle. The perimeter of the window is 8 metre.

(a) Express the front surface area of the window in terms of x . [4 marks]

(b) Find the length (in metre) of the window when the front surface area is maximum. [6 marks]

END OF QUESTION PAPER

Answers

SECTION A

1. (a) -4 (b) $-\frac{\sqrt{2}}{5}$

2. (a) $\frac{dy}{dx} = x^{\sec x} \sec x \left(\frac{1}{x} + \tan x \ln x \right)$ (b) $-\frac{4x}{\sqrt{4x^2 + 1}} \operatorname{cosec}^2(4x^2 + 1)^{\frac{1}{2}}$

3. $(-2, 23)$ is maximum point, $\left(3, -\frac{79}{2}\right)$ is minimum point

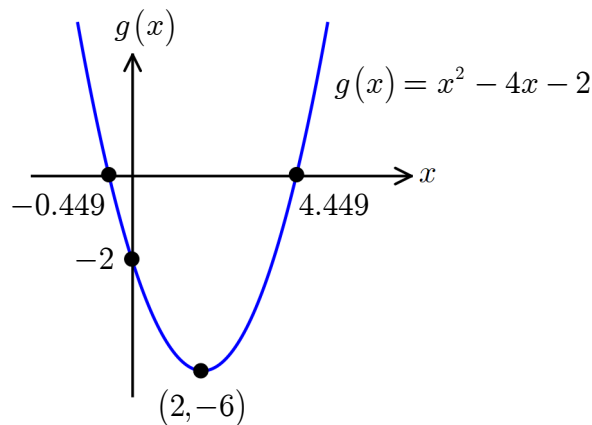
SECTION B

1. $\left| \bar{z} - \frac{1}{\bar{z}} \right| = 2, \alpha + \beta i = \frac{56}{25} - \frac{192}{25}i$

2. (a) $x = 8, x = \frac{1}{32}$ (b) $[-2, -1] \cup [1, 2]$

3. (a) $g(x) = (x - 2)^2 - 6$ (b) $D_g = (-\infty, \infty), R_g = [-6, \infty)$

(c)



4. (a) $f(x) = \frac{x+1}{2x+1}, f\left(\frac{1}{2}\right) = \frac{3}{4}$ (b) $x \neq 0, -1, -\frac{1}{2}$

5. (a) $a = -\frac{3}{2}, b = -6$ (b) VA: $x = -1, x = 1$, HA: $y = -1, y = 1$

6. (a) $a = 5, b = -5$ (b) $\frac{d^2y}{dx^2} = -\frac{3}{4}$

7. (a) $A = (-2\pi - 8)x^2 + 16x$ (b) 2.2404 m

SECTION A [25 MARKS]

This section consists of **3** questions. Answer **all** questions.

1. (a) Find the value of m if $\lim_{x \rightarrow 0} \frac{mx + 3x^2}{4x - 8x^2} = 3$. [3 marks]
- (b) Evaluate
- (i) $\lim_{x \rightarrow 0} \frac{x}{\sqrt{3-x} - \sqrt{3}}$ [3 marks]
- (ii) $\lim_{x \rightarrow 3^+} \frac{x^2 - 5x + 6}{|3 - x|}$ [3 marks]
2. Given $f(x) = \frac{3x + 1}{(x - 1)(x^2 + 1)}$. By expressing $f(x)$ in partial fractions, find $f'(x)$ and hence, evaluate $f'(3)$. [9 marks]
3. The curve $f(x) = \frac{px + r}{x^2 - 3x}$ has two stationary points at $(-1, -1)$ and at point H . Determine the values of p and r . State the coordinate of H . [7 marks]

SECTION B [75 MARKS]

This section consists of **7** questions. Answer **all** questions.

1. Find $\sqrt{8+6i}$ in the form of $a+bi$, where $a, b \in \mathbb{R}$. [6 marks]
2. (a) Solve $\log_3 x - 1 \leq 6 \log_x 3$. [7 marks]
(b) Solve the inequality $|x-1| \leq 2|x+3|$. [5 marks]
3. Consider functions of $f(x) = (x-2)^2 + 1$, $x > 2$ and $g(x) = \ln(x+1)$, $x > 0$.
(a) Find f^{-1} and g^{-1} . State the domain and range for each of the inverse function. [8 marks]
(b) Obtain $(g \circ f)(x)$. [2 marks]
4. Given that $g(x) = e^{2x} - 3$ and $f \circ g(x) = 2x$. Find $f(x)$. Hence, find $f^{-1}(x)$. State its domain and range. [8 marks]
5. (a) A function f is defined by $f(x) = \frac{1+x}{\sqrt{x^2-4}}$.
(i) Find the vertical asymptotes. [3 marks]
(ii) Determine $\lim_{x \rightarrow +\infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.
Hence, state the horizontal asymptote(s). [4 marks]
- (b) Function f is given by $f(x) = \begin{cases} 1-ax^2, & x \leq -2, \\ 2x+a, & x > -2, \end{cases}$ is continuous for all values of x .
(i) Find the value of the constant a . [3 marks]
(ii) Find the $\lim_{x \rightarrow -2} \frac{f(x) - f(-2)}{x+2}$ if it exists. [4 marks]

6. (a) The parametric equations of a curve are given by $x = e^{2t+1}$ and $y = e^{-(2t-1)}$.

(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at point (e^3, e^{-1}) . [6 marks]

(ii) Given $z = x^2 - xy$. Express z in terms of t and find $\frac{dz}{dt}$.

Hence, deduce the set value of t such that $\frac{dz}{dt}$ is positive. [5 marks]

(b) Given $y = \tan\left(\frac{\pi}{4} + x\right)$, show that $\frac{d^2y}{dx^2} - 2y \frac{dy}{dx} = 0$. [4 marks]

7. Oil is poured into a hemisphere bowl of radius 5 cm at the rate of $4\pi \text{ cm}^3/\text{s}$. The volume,

$V \text{ cm}^3$, of the oil in the bowl when its depth is $h \text{ cm}$ is given by $V = \pi\left(5h^2 - \frac{1}{3}h^3\right)$.

At the instant when the depth of the oil in the bowl is 2 cm, find

(a) the rate of change of h , [3 marks]

(b) the rate of change of the radius of the oil's surface area. [7 marks]

END OF QUESTION PAPER

Answers

SECTION A

1. (a) $m = 12$ (b) $-2\sqrt{3}$ (c) 1
2. $f'(x) = -\frac{2}{(x-1)^2} + \frac{2x^2 - 2x - 2}{(x^2 + 1)^2}$, $f'(3) = -\frac{2}{5}$
3. $r = 1$, $p = 5$, $H = \left(\frac{3}{5}, -\frac{25}{9}\right)$

SECTION B

1. $z = 1 + 3i$, $z = -1 - 3i$
2. (a) $\left\{x : x \leq \frac{1}{9} \text{ or } 1 < x \leq 27\right\}$ (b) $(-\infty, -7] \cup \left[\frac{-5}{3}, \infty\right)$
3. (a) $f^{-1}(x) = \sqrt{x-1} + 2$, $D_{f^{-1}} = (1, \infty)$, $R_{f^{-1}} = (2, \infty)$
 $g^{-1}(x) = e^x - 1$, $D_{g^{-1}} = (0, \infty)$, $R_{g^{-1}} = (0, \infty)$
(b) $(g \circ f)(x) = \ln(x^2 - 4x + 6)$
4. $f(x) = \ln(x+3)$, $f^{-1}(x) = e^x - 3$, $D_{f^{-1}} = \mathbb{R}$, $R_{f^{-1}} = (-3, \infty)$
5. (a)(i) VA: $x = -2$, $x = 2$ (ii) HA: $y = -1$, $y = 1$
(b)(i) $a = 1$ (ii) the limit does not exist
6. (a)(i) $\left.\frac{dy}{dx}\right|_{t=1} = -e^{-4}$, $\left.\frac{d^2y}{dx^2}\right|_{t=1} = 2e^{-7}$ (ii) $\{t : t \in \mathbb{R}\}$ (b) *DIY*
7. (a) $\frac{dh}{dt} = \frac{1}{4} \text{ cm/s}$ (b) $\frac{3}{16} \text{ cm/s}$